

BIOS vs. UEFI Comparison

ITEP 414 – System Administration and Maintenance | Week 3 Portfolio Project

Comparison Table

Criteria	BIOS	UEFI
Definition	Legacy firmware interface embedded in the motherboard chip	Modern firmware interface, successor to BIOS
Boot Process	Sequential; reads the boot sector from the MBR	Faster pre-boot environment with its own drivers
Maximum Disk Support	~2.2 TB (limited by MBR)	Over 9 zettabytes
Partition Style	MBR (Master Boot Record); max 4 primary partitions	GPT (GUID Partition Table); up to 128 partitions
Security Features	Minimal / none; no Secure Boot	Secure Boot, network authentication support
Speed	Slower (16-bit mode, sequential initialization)	Faster (32/64-bit mode, parallel initialization)
Advantages	Simple, long-proven, compatible with very old hardware	Large disk support, mouse-driven GUI, stronger security, network boot
Disadvantages	Limited disk size, no GUI, weak security model	More complex; possible compatibility issues with very old operating systems
Modern Usage	Largely phased out; found only in legacy systems	Standard on virtually all systems shipped since ~2012

Why UEFI Has Largely Replaced BIOS in Modern Computers

BIOS was used for more than thirty years as the first thing that ran when a computer powered on. However, due to its 16-bit architecture, it is limited to very small memory access and a maximum disk size of about 2.2 TB — no longer sufficient for modern storage devices that now reach into the terabytes.

Because of this, UEFI arrived as a more capable and flexible replacement. Using the GPT partition scheme, UEFI can support very large disks (up to zettabytes) and up to 128 partitions, compared to only four primary partitions under BIOS. In addition, UEFI offers Secure Boot, a security feature that prevents unauthorized or malicious software from loading during startup — an important protection against bootkits and rootkits.

UEFI also provides a graphical user interface that supports a mouse, making it noticeably easier to use than the text-only, keyboard-driven interface of BIOS. It also boots faster due to its 32-bit or 64-bit operating mode, and supports network boot and remote diagnostics — a significant benefit for IT administrators managing multiple servers remotely.

For all these reasons, most modern operating systems — including Windows 11 and recent Linux distributions such as Ubuntu Server — now require or strongly recommend UEFI as the default firmware standard.